

Document Control

- Document Type: Business Requirements Document (BRD)
- System Name: Application

Executive Summary

- This BRD defines the business requirements for ingesting customer and order CSV data into Delta tables.
- Focuses on data cleansing, enrichment, SCD Type 2 change handling, and daily spend aggregation.
- The pipeline is implemented using Delta Live Tables.
- System Name: Application

Business Objectives

- Load customer and order source CSVs into Delta tables.
- Cleanse datasets by removing "Null"/Null literals and duplicates.
- Enrich order data with a computed TotalAmount field.
- Join customer and order data to create an ordersummary table, applying SCD Type 2 logic.
- Create a customeraggregatespend table for daily customer spend totals.
- End-to-end data pipeline orchestrated using Delta Live Tables.

In-Scope

- Ingestion from specified file paths for customer and order CSVs.
- Delta table creation: customer, order, ordersummary, customeraggregatespend.
- Data cleansing operations for both datasets.
- Adding computed TotalAmount field in order.
- Join operation and SCD Type 2 logic in ordersummary.
- Aggregating spend in customeraggregatespend.
- Implementation via Delta Live Tables.

Out-of-Scope

- Data sources and fields outside the provided JSON requirements.
- Security, access controls, SLAs, monitoring, alerting mechanisms.
- Performance tuning and scaling.
- Scheduling and orchestration details not tied to "Delta Live Tables."
- Additional data quality metrics or reporting/analytics layers.

Stakeholders

- Stakeholder details not specified in requirements.

Business Requirements

BR-APP-001 — Source Data Ingestion

- Description: Ingest customer and order data from specified CSV file locations into Delta tables.
- Business Value: Enables unified, structured data storage for downstream processing.
- Priority: High

BR-APP-002 — Data Schema Enforcement

- Description: Enforce target schema on customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId) tables.
- Business Value: Ensures data consistency for later processing steps.
- Priority: High

BR-APP-003 — Data Cleansing Operations

- Description: Remove rows containing "Null"/Null literals and any duplicate records from both datasets.
- Business Value: Improves downstream data quality and integrity.
- Priority: High

BR-APP-004 — Computed Column Creation

- Description: Compute a TotalAmount field as PricePerUnit × Qty in the order table.
- Business Value: Provides essential measure for aggregation and business insight.
- Priority: High

BR-APP-005 — Ordersummary Table Creation

- Description: Create an ordersummary table by joining customer and order on CustId, applying SCD Type 2 logic for customer changes.
- Business Value: Maintains a complete view and history of customer-order relationships.
- Priority: High

BR-APP-006 — SCD Type 2 Change Handling

- Description: For customer changes, close old records, insert new active records, and update StartDate/EndDate to retain change history.
- Business Value: Enables historical tracking for regulatory and analytical use.
- Priority: High

BR-APP-007 — Aggregation Table Creation

- Description: Aggregate ordersummary data by Name and Date to create customeraggregatespend with total daily spend.
- Business Value: Delivers actionable business metrics and supports analytics.
- Priority: High

BR-APP-008 — Pipeline Implementation

- Description: Implement all steps using Delta Live Tables.
- Business Value: Simplifies orchestration and ensures consistency.
- Priority: Medium

Assumptions & Constraints

- Only actions and attributes explicitly listed in requirements are assumed.
- Delta Live Tables is the prescribed solution for implementation.
- StartDate, EndDate, Active flag fields are implied in SCD Type 2 logic but detailed field definitions are not specified.

Success Metrics

- 100% of ingested rows validated against schema and cleansing rules.
- Accurate TotalAmount calculation in all order records.
- Ordersummary table reflects full SCD Type 2 history for all customer changes.
- Customeraggregatespend reflects correct daily aggregations by Name.
- All pipeline steps executed successfully via Delta Live Tables.

Risks & Mitigations

- Missing explicit data types or field definitions for SCD tracking fields
- Mitigation: Consult data model team or clarify with stakeholders.
- No error handling or monitoring processes defined
- Mitigation: Propose adding standard data pipeline monitoring as a future enhancement.
- Unknown expected data volume or performance SLAs
- Mitigation: Include scalability evaluation after initial pipeline deployment.

Approval

- To be completed by project sponsors and key stakeholders upon review.

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
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customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Source for Delta table customer
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Source for Delta table order

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Raw	customer	CustId, Name, EmailId, Region	Loaded
	Raw	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Loaded
	Processed	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	SCD Type 2
spend	Aggregate	customeraggregatespend	Name, TotalAmount, Date	Aggregated

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Ingest	customerdata	customer	CSV to Delta table
2	Ingest	orderdata	order	CSV to Delta table
3	Cleanse	customer, order	cleansed tables	Remove "Null"/Null literals and duplicates
4	Compute	order	order	TotalAmount = PricePerUnit × Qty
5	Join	customer, order	ordersummary	Join on CustId and apply SCD Type 2
6	Aggregate	ordersummary	customeraggregatespend	Sum TotalAmount by Name and Date

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-APP-001	Cleansing	Remove "Null"/Null literals and duplicates from all tables	Requirements
BR-APP-002	Computation	TotalAmount = PricePerUnit × Qty in order table	Requirements
BR-APP-003	Join	Join customer and order on CustId	Requirements
BR-APP-004	SCD Type 2	Maintain historical customer changes in ordersummary	Requirements
BR-APP-005	Aggregation	Group and sum TotalAmount by Name, Date	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customerdata into customer Delta table	Source available
2	Ingest orderdata into order Delta table	Source available
3	Cleanse data (remove Nulls, duplicates)	Step 1, Step 2
4	Compute TotalAmount in order	Step 3
5	Join customer and order, create ordersummary with SCD Type 2	Step 4
6	Aggregate spend by Name/Date, create customeraggregatespend	Step 5

Figure 1: End-to-End Data Pipeline Flow

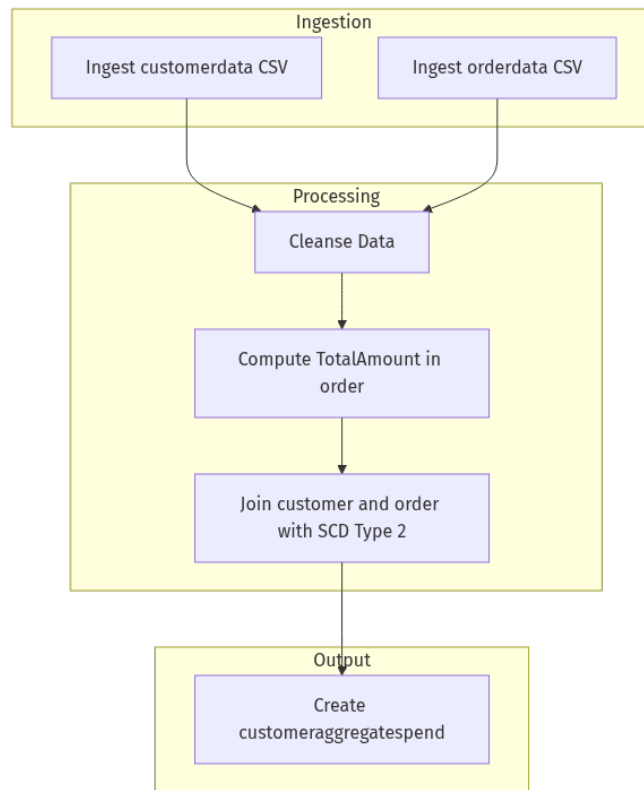


Figure 2: SCD Type 2 Processing Logic

